Standard and Long-Acting Depot Neuroleptics in Chronic Schizophrenics

An 18-Month Open Multicentric Study

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• The overall objective of this 18-month open study was to compare standard neuroleptics and long-acting depot neuroleptics following the current psychiatric practice in order to determine the best therapy. Thirty French psychiatrists from 15 different wards participated in this experiment. One hundred eighty-one chronic schizophrenic patients were randomly assigned to receive one of the following three treatments: standard neuroleptics, pipotiazine palmitate, or fluphenazine decanoate. Criteria used for evaluation were an overall clinical evaluation by a psychiatrist, a Brief Psychiatric Rating Scale, and a Nurse's Observation Scale for Inpatient Evaluation. No significant difference ($P > .05$) was observed between the three groups in drug effectiveness or tolerance.

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The effectiveness of standard neuroleptics (SN) has long been confirmed by controlled studies as a preventive treatment of schizophrenic relapses.[1] Long-acting depot neuroleptics (DN) have a significant superiority when compared with placebos in the same indications.[2-4]

A large number of relapses, during SN treatment, can be attributed to patient's interruption of therapy.[5] It is noted that 30% to 40% of the patients do not take their oral medication during long-term treatment.[6] One of the principal advantages of DN is to eliminate this problem and thus reduce the number of relapses due to treatment abandonment.[7,8]

However, recent studies[9] have shown that 30% of patients treated by DN relapse as though the drug effectiveness decreased after a few months' time. Contrary to this last finding, an obvious superiority was seen for DN over SN in the prevention of relapses following a literature search by Docherty et al.[10]

The aim of our study was to test two different methods of drug administration in severe chronic schizophrenics over a period of 18 months. The two treatments, SN and DN, were administered following the current psychiatric practice. These two therapeutic methods must be compared in both their practical advantages and disadvantages and therefore only an open study could be used.[11,12] Concerning SN, psychiatrists have the choice of the drug(s), its (their) dosage, and can substitute drugs during the course of patient management. Under these conditions, a double-blind study could not be achieved. The regular taking of SN was not controlled because of known patient noncompliance to daily ingestion of medication.[5,6] Furthermore, a double-blind study must include a double placebo and the depot placebo would have compelled the psychiatrist to follow an unusual course of SN treatment. Two different DN (pipotiazine palmitate and fluphenazine decanoate) were chosen as they were thought to have different therapeutic effects. Thus, three drug groups were studied: one SN and two DN.

Contrary to many studies, the evaluation criteria used in this study were not the number of relapses observed while under therapy, but rather a subjective clinical analysis by a psychiatrist and measurements by rating scales.

PATIENTS AND METHODS

The Patients

Male and female chronic schizophrenic patients (ages 21 to 45 years), as defined by official French classification of mental illness,[13] were included in this experiment, that is, simple, hebe-

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phrenic, catatonic, paranoid, schizoaffective, and residual schizophrenias. An illness duration of more than three years was required to include a patient in this study. For each patient, the experimenters were required to estimate that a maintenance neuroleptic chemotherapy for at least 18 months was necessary. Most were inpatients.

Patients were excluded for the following reasons: a somatic illness, a lobotomy, a Sakel cure during the three previous years, or a known intolerance to neuroleptics.

Centers and Experimenters

All French hospital psychiatrists were asked to participate. One hundred five positive responses were received. After the details of the protocol[12] were explained, 31 possible wards remained; among these, 15 were randomly chosen.

In each center, two psychiatrists participated in the experiment, examined separately all the patients who were likely to enter the experiment, verified separately the criteria of entering the experiment, and divided the patients between them. If for some unforeseen reason one psychiatrist became unavailable, the other then took charge of all the patients in that center.

Randomization

The available number of patients for each center was limited to a minimum of 12 and a maximum of 18. For each center, a balanced randomization was prepared placing each patient in one of the following groups: (1) SN group: The neuroleptics were chosen and modified with all the freedom according to usual practice (free choice of dosage, one or more SN). All long-acting neuroleptics and any kind of noncommercialized drugs were forbidden. (2) Fluphenazine decanoate group: The depot-neuroleptic was administered intramuscularly at doses and intervals chosen by the psychiatrist in charge. All the other neuroleptics were forbidden, except for methotrimeprazine in conditions defined below. If the prescription of other neuroleptics became necessary, this ended the experiment for this patient. (3) Pipotiazine palmitate group: The conditions were the same as those of the second group.

To control excessive anxiety, the prescription of up to 100 mg/day of methotrimeprazine was authorized for a maximum period of one month in groups 2 and 3. In all the groups, the prescription of antiparkinsonian drugs and of minor tranquilizers and hypnotics was authorized. In case of a depressive state, the prescription of an antidepressant for a short period of time (maximum one month) was authorized.

After all these conditions were met, the patient was allowed to enter the experiment, and was randomly placed in one of the three groups. To avoid a problem with intolerance, patients placed in groups 2 and 3 were tested with the corresponding oral neuroleptic (pipotiazine or fluphenazine) for 15 days.

Evaluation Criteria

The evaluation of the patients was accomplished in two different stages: initially and the 18th month.

Initially, as soon as the decision "good for experiment" was taken by the two psychiatrists of each center, a Brief Psychiatric Rating Scale (BPRS) was filled out for each patient and a permanent member of the nursing team filled out a Nurse's Observation Scale for Inpatient Evaluation (NOSIE). The psychiatrist was also required to take a short history of the illness and to indicate from a preestablished list which positive and/or negative symptoms he hoped would be improved.

At the 18th month, the following was filled out: (1) an overall clinical evaluation, using a five-point scale (aggravation, no change, symptomatic or transitory improvement, good improvement, complete remission); (2) an evaluation of any modification of the patient's adaptation to the "milieu," using a three-point scale (aggravated, no change, improved); (3) a BPRS; (4) an appreciation of the tolerance using a three-point scale (poor, few problems, good) and an analytic inventory of the neurologic side effects (rated 0 to 3); and (5) a NOSIE, filled out by the same nurse.

Statistical Analysis

In order to compare the three groups, we used an analysis of variance for the quantitative variables. The qualitative variables were first regrouped when necessary to avoid very small sample sizes. Then the usual or corrected χ^2 test or the exact Fisher's test were used, depending on the sample size (for example, Fisher's test for the smallest). In all cases, the test was declared nonsignificant (NS) when $P > .05$. The values of P are indicated for significant tests.

RESULTS

Baseline Comparisons of the Three Groups

A homogeneous distribution was fortunately observed in each of the three groups in all the following characteristics: sex, age, diagnosis, sickness duration, initial symptomatology (acute or stable state), positive or negative symptoms, total average score for the BPRS and the NOSIE (Table 1). The global clinical evaluation of the illness severity by the nurses (using the seven-point scale from item 31 of the NOSIE: "Based on your clinical experience, what is the present severity of the patient's sickness?") also showed an identical distribution for the three groups.

Analysis of Dropouts

Out of a starting sample of 181 patients, 39 dropouts were observed. The number of interruptions with the corresponding reasons are indicated in Table 2. The 18 dropouts due to independent causes and those lost to follow-up were not considered in the analysis. This decision was possible because there was a homogeneous distribution in the three groups (χ^2 NS). Finally, there were 163 patients considered. The 21 patients that left the trial for the three other reasons were all attributed to treatment failures. For 11 of these dropouts, their conditions were evaluated when they left the experiment, using the procedures that were to be used at the 18-month examination. These evaluations were considered in the global analysis. For the other patients with missing data an arbitrary decision was made as follows: (1) concerning the overall clinical evaluation, it was decided to consider their state as aggravated; (2) concerning BPRS and NOSIE, a value was chosen for the difference between the total score at 18 months and the initial total score, by calculation (grossly corresponding to the 5% worst patients).

Characteristics of the Treatments

In the SN group, the average number of different neuroleptics prescribed per patient during the period of the experiment was 2.3. The average dose of 90 mg every 25 days was administered for the pipotiazine palmitate group, and 88 mg every 22 days for the fluphenazine decanoate group.

Concerning the associated authorized treatments, there was no significant difference in the use of antiparkinsonian and antidepressant drugs (Table 3).

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Table 1.—Baseline Comparisons of the Three Groups After Randomization

	Patient Group			
	SN (N = 63)	Pipotiazine Palmitate (N = 61)	Fluphenazine Decanoate (N = 57)	P
Sex				
M	40	41	36	NS*
F	23	20	21	
Age, yr				
20-29	15	21	14	
30-39	27	21	23	NS*
> 40	21	19	20	
Diagnosis: Schizophrenia				
Paranoid	34	29	33	
Hebephrenic	12	12	13	NS*
Other	17	20	11	
Sickness duration, yr				
3-5	13	13	10	
5-10	15	21	10	NS*
>10	35	27	37	
Initial symptomatology				
Acute state (N = 58)	17	20	21	NS*
Stable state (N = 123)	46	41	36	
Positive symptoms				
Delusions	40	41	41	
Hallucinations	25	20	30	NS*
Agitation	28	16	19	
Negative symptoms				
Autism	39	41	25	
Apragmatism	48	49	42	NS*
"Milieu" maladaptation	46	46	40	
BPRS				
Total average score	61.08	58.02	60.33	NS†
SD	±12.98	±11.32	±14.80	
NOSIE				
Total average score	152.81	157.46	152.25	NS†
SD	±18.55	±14.04	±16.97	

*χ^2.
†Analysis of variance.

Table 2.—Classification of Dropouts

	No. of Patients		
	SN	Pipotiazine Palmitate	Fluphenazine Decanoate
Lost to follow-up	2	2	3
Causes certainly independent of treatment	4	4	3
Drug intolerance	0	1	3
Drug ineffectiveness	4	8	4
Depressive state	0	0	1

Table 3.—Associated Treatments

	No. of Patients			
	SN (N = 63)	Pipotiazine Palmitate (N = 61)	Fluphenazine Decanoate (N = 57)	P
Antiparkinsonian drugs				
Never used	14	19	11	
Used occasionally	14	8	8	NS*
Always used	35	34	38	
Antidepressant drugs				
Used occasionally	5	4	5	NS*
Never used	58	57	52	

*χ^2.

Table 4.—Overall Clinical Evaluation

	No. of Patients		
	SN	Pipotiazine Palmitate	Fluphenazine Decanoate
Aggravation	2 (6)*	2 (10)*	4 (10)*
No change	22	13	9
Transitory symptomatic improvement	17	17	18
Good improvement	9	11	10
Complete remission	3	4	4

*The numbers in parentheses are those obtained after introduction (as aggravated) of dropouts due to drug intolerance or ineffectiveness.

Overall Clinical Evaluation

The results are indicated in Table 4. There was no significant difference between the three groups before or after introduction of missing data (χ^2 NS).

From the overall clinical evaluation data, we also attempted to see if there was a significant difference between the three treatments according to the initial symptomatology (acute or stable state) (Table 5). No significant difference was seen (χ^2 NS).

We also tested for a relationship between initial symptomatology (acute or stable state) and patient improvement in each of the three treatments separately. The comparison between improved patients and other patients is indicated in Table 6.

Adaptation to the 'Milieu'

The results at 18 months on the adaptation to the "milieu" (Table 7) take into consideration only the patients that reached the end of the experiment and those for whom we obtained data when they dropped out.

BPRS

To evaluate the BPRS results, we calculated the differences between the total score at 18 months and before treatment for each patient. It was noted that even though there was no significant difference between the three groups, there was a tendency toward improvement. This is shown by a negative difference in Table 8.

NOSIE

Besides the total average score (Table 9), we also studied in more detail the answer to item 32 of the NOSIE at 18 months: "If you compare his present state with that at the beginning of the study, what is the change that has occurred?" (Table 10).

Side Effects

We asked the psychiatrists to evaluate in a global way the tolerance to treatment. Results are indicated in Table 11. In the case of moderate or poor tolerance, the evaluators were asked to indicate from a proposed list the nature and the degree of the observed symptomatology. The results

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Table 5.—Overall Clinical Evaluation Based on Initial Symptomatology

	No. of Patients		
	SN	Pipotiazine Palmitate	Fluphenazine Decanoate
Stable state			
Aggravation	4	5	7
No change	20	12	7
Transitory symptomatic improvement	15	10	12
Good improvement	6	9	5
Complete remission	0	0	1
Acute state			
Aggravation	2	5	3
No change	2	1	2
Transitory symptomatic improvement	2	7	6
Good improvement	3	2	5
Complete remission	3	4	3

Table 6.—Effects of Different Treatments According to Initial State

	No. of Patients		
	Improved Patients*	Other Patients†	P
SN			
Stable state	6	39	$P < .01$‡
Acute state	6	6	
Pipotiazine palmitate			
Stable state	9	27	NS‡
Acute state	6	13	
Fluphenazine decanoate			
Stable state	6	26	$P < .05$‡
Acute state	8	11	

*Good improvement and complete remission.
†Transitory symptomatic improvement, no change or aggravation as measured by the overall clinical evaluation.
‡Fisher's exact test.

Table 7.—Adaptation to the 'Milieu': 18 Months' Results

	No. of Patients			
	SN	Pipotiazine Palmitate	Fluphenazine Decanoate	P
Aggravated	4	0	1	
No change	30	25	19	NS*
Improved	19	22	25	

*χ^2.

Table 8.—Difference Between BPRS Total Average at 18 Months and Before Treatment

	Patient Group			
	SN	Pipotiazine Palmitate	Fluphenazine Decanoate	P
	(N = 53)	(N = 47)	(N = 45)	
Difference	−11.55	−13.96	−13.51	
				NS*
SD	± 2.31	± 2.56	± 2.83	
Results With Calculated Values (for Missing Data)				
	(N = 57)	(N = 55)	(N = 51)	
Difference	−10.32	−10.01	−10.52	
				NS*
SD	± 2.24	± 2.56	± 2.77	

*Analysis of variance.

Table 9.—Difference Between NOSIE Total Average Score at 18 Months and Before Treatment*

	Patient Group			
	SN	Pipotiazine Palmitate	Fluphenazine Decanoate	P
	(N = 52)	(N = 51)	(N = 48)	
Difference	+3.61	+6.64	+8.80	
				NS*
SD	±2.71	±2.02	±2.71	

*Analysis of variance.

Table 10.—Answer to Item 32 of NOSIE

	No. of Patients			
	SN	Pipotiazine Palmitate	Fluphenazine Decanoate	P
Aggravated	3	1	1	
No change	22	16	11	
Slightly improved	11	14	19	NS*
Moderately improved	5	5	1	
Greatly improved	5	10	10	

*χ^2.

show that there was only a difference (although nonsignificant) between treatments concerning tardive dyskinesia: three in the SN group, three in the pipotiazine palmitate group, and none in the fluphenazine decanoate group. Finally, we observed four dropouts because of treatment intolerance (Table 12).

COMMENTS

This study was undertaken with particularly severe, chronic schizophrenic patients, more than 50% of whom had a sickness duration of greater than ten years.

Of the criteria used for assessment, we felt that the overall clinical evaluation by the psychiatrist was the best for noting any change over 18 months. We also used the BPRS, but because most of the patients were severe, chronic schizophrenics we did not anticipate seeing any great modification with this scale. The NOSIE was another scale employed. It is noted that some NOSIE scales were not completed, either because the patient was ambulatory initially or because he became an outpatient during the trial. Although currently noted in the literature, we did not take into consideration the following evaluation criteria of effectiveness: the length of hospitalization, the number of relapses, or the number of hospitalizations. The sample obtained was too rich in severe, chronic inpatients, for whom the level of therapeutic expectation was only moderate at the beginning of the experiment.

The number of dropouts was moderate. We observed seven subjects out of 181 lost to follow-up, equally distributed in the three groups.

It is important to note that approximately 8% of the subjects in each group developed depressive symptoms and required the prescription of antidepressants. Furthermore the fact that antiparkinsonian drugs were always prescribed for a majority of the patients does not neces-

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Table 11.—Treatment Tolerance

	No. of Patients			
	SN	Pipotiazine Palmitate	Fluphenazine Decanoate	P
Poor	1	1	3	
Moderate	14	9	5	NS*
Good	39	38	38	

*χ^2.

Table 12.—Dropouts Secondary to Drug Intolerance

Treatment	Period of Treatment, mo	Reason
Fluphenazine decanoate (N = 3)	1	Severe parkinsonian syndrome
	2	Apathy and motor retardation
	11	Akathisia, tasikinesia
Pipotiazine palmitate (N = 1)	5	Apathy, slight parkinsonian syndrome

sarily signify a great frequency of parkinsonian syndromes, but rather a systematic French psychiatric practice.

The objective of this study was to test two therapeutic methods (SN vs DN), in order to evaluate which was superior for the treatment of chronic schizophrenia. After 18 months and with naturalistic conditions, no real difference appeared between the three groups compared. This lack of difference was observed not only in the different evaluation criteria used (overall clinical evaluation, BPRS, NOSIE), but also in the need to prescribe associated drugs (antiparkinsonians, antidepressants).

Whatever the treatment used, we did not note a dramatic change in the patients' condition over an 18-month period in a group of 163 schizophrenics followed up in an identical way.

However, after 18 months, 21 patients of the 113 with a stable state were greatly improved. These 21 patients received an evaluation of either "good improvement" or "complete remission" by the overall clinical evaluation. A greater number of acute state patients (20 out of 50) showed the same improvement. Thus, we confirmed the known equivalency of the two treatments (SN and DN) for stable state patients and also showed that the treatment of a chronic schizophrenic's acute attack with SN or DN (simpler and easier) would give the same chance of success.

Globally, we can say that the results obtained with the different measuring criteria, whether they be utilized by the psychiatrist or by the nursing team, are very similar. It is interesting to note that for this type of patient, the overall clinical evaluation by the psychiatrist, the total average score of the BPRS and of the noise, and the global evaluation by the nurse (item 32 of the NOSIE) all varied in the same way, that is, a tendency toward amelioration. A detailed correlation study for each patient on all these points will be the subject of a future article.

Two questions still remain to be answered. We will analyze these in more detail at a later date with the aid of these results: (1) Is there a difference in treatment effectiveness according to the clinical type of schizophrenia? (2) Is there a difference between treatments based on the diverse factors of BPRS and NOSIE?

CONCLUSION

The conclusions we draw from these data are (1) there was no statistical difference ($P > .05$) in the patients' clinical course (based on the overall clinical evaluation, BPRS, and NOSIE) between the groups, no matter whether the treatment used was standard neuroleptics (SN) or long-acting depot neuroleptics (DN—pipotiazine palmitate or fluphenazine decanoate); (2) the course of the illness was on the average favorable for all treatments used; (3) the lack of differences observed between the treatments was consistent no matter whether it was begun for a patient in an acute or in a stable state; (4) the average dosage of DN used by the French psychiatrists was superior to the common dosage prescribed by American psychiatrists; (5) although our study was realized under conditions as close as possible to the habitual treatment of chronic schizophrenics, the number of dropouts observed was inferior to the usual number stated in the literature; and (6) no difference in patient tolerance to SN, to pipotiazine palmitate, and to fluphenazine decanoate was noted.

The participating psychiatrists were as follows: P. Angel, MD; G. Audoux, MD; Y. Bastie, MD; P. Bobet, MD; J. M. Boissenin, MD; G. Bollotte, MD; S. Brion, MD; G. Charriot, MD; G. Clerc, MD; F. Coloni, MD; M. J. Cottereau, MD; M. Cottin, MD; F. Finance, MD; A. Gisselmann, MD; O. Grandmontagne, MD; L. Grimberg, MD; R. Guérin, MD; C. Lambic, MD; J. P. Legendre, MD; L. Levy, MD; H. Lôo, MD; A. Marin, MD; P. Resche-Rigon, MD; R. Ropert, MD; M. Sales, MD; P. Sevestre, MD; L. Singer, MD; J. L. Vallet, MD; E. Zarifian, MD; and M. P. Zuber, MD.

C. Degos, MD; G. Dordain, MD; and C. Rouquès, MD, participated in the protocol redaction, and S. Lancrenon performed the statistical calculations. J. Dubois provided secretarial help.

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